

Question

Write a Java program to write text to a file, read it, and count the occurrences of "pe" and "pi" without case sensitivity.

Steps for Execution

1. Save the code as `FileHandling.java`.
2. Compile the program.
3. Run `FileHandling`.
4. The program creates and reads `sample.txt`, then prints both counts.

Algorithm

1. Store the given text in a string.
2. Write the text to `sample.txt`.
3. Read the file one character at a time.
4. Convert the text to lowercase.
5. Count each occurrence of "pe" and "pi".
6. Display the counts.

UML Class Diagram

```
+-----+
|           FileHandling           |
+-----+
| + main(args: String[]): void    |
|   writes and reads sample.txt   |
|   counts "pe" and "pi"         |
+-----+
```

CODE: SRC

FileHandling.java

```

import java.io.FileWriter;
import java.io.FileReader;
import java.io.IOException;

public class FileHandling {

    public static void main(String[] args) {

        String text =
            "Peter Piper picked a peck of pickled peppers\n" +
            "A peck of pickled peppers Peter Piper picked\n" +
            "If Peter Piper picked a peck of pickled peppers\n" +
            "Where's the peck of pickled peppers Peter Piper picked?";

        try {
            // Writing data into the file
            FileWriter writer = new FileWriter("sample.txt");
            writer.write(text);
            writer.close();

            // Reading data from the file
            FileReader reader = new FileReader("sample.txt");

            String content = "";
            int ch;

            while ((ch = reader.read()) != -1) {
                content += (char) ch;
            }

            reader.close();

            // Convert to lowercase for counting
            content = content.toLowerCase();

            int peCount = 0;
            int piCount = 0;

            // Count "pe"
            for (int i = 0; i < content.length() - 1; i++) {
                if (content.substring(i, i + 2).equals("pe")) {
                    peCount++;
                }
            }
        }
    }
}

```

```

        if (content.substring(i, i + 2).equals("pi")) {
            piCount++;
        }
    }

    System.out.println("'pe' - no of occurrences - " + peCount);
    System.out.println("'pi' - no of occurrences - " + piCount);

} catch (IOException e) {
    System.out.println("An error occurred.");
}
}
}

```

Compilation and Execution

```

javac FileHandling.java
java FileHandling

```

Output

```

'pe' - no of occurrences - 20
'pi' - no of occurrences - 12

```